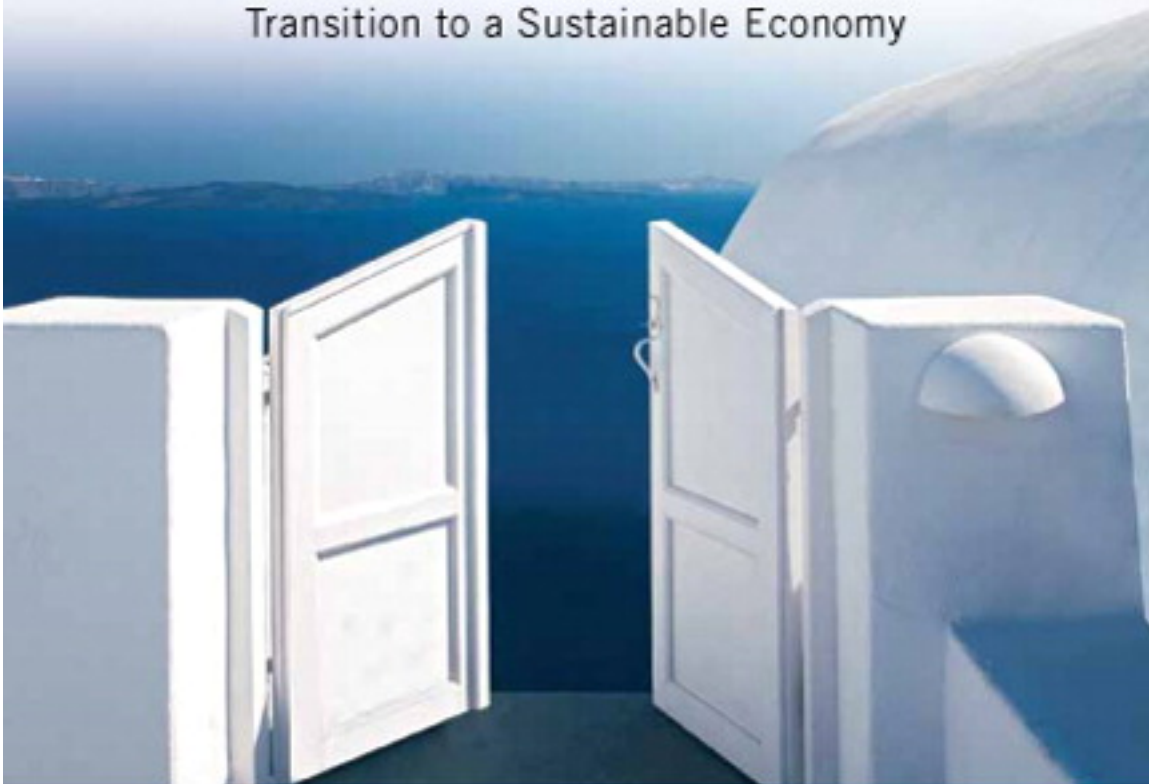




Sustainable Venturing

Entrepreneurial Opportunity in the
Transition to a Sustainable Economy



Thomas J. Dean

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CHAPTER

INSTITUTIONAL STRATEGIES FOR SUSTAINABLE VENTURING

INTRODUCTION

In the last chapter, we discussed economic systems and their role in motivating, or dissuading entrepreneurial action. The basic contention was that, with respect to natural and environmental resources, conditions of market failures could prevent entrepreneurs from implementing businesses that enhance ecological or social sustainability. In addition, we noted that conditions of market failure can be, and are being, altered through the processes of eco-institutional convergence. We also suggested that market failures are, at their essence, sources of entrepreneurial opportunity. In other words, win-lose conditions can be translated into win-win conditions through entrepreneurial action.

Environmental entrepreneurs engage in the processes of *opportunity recognition* and/or *opportunity creation* with respect to the institutional changes associated with eco-institutional convergence.¹ *Opportunity recognition* refers to the ability of entrepreneurs to perceive market gaps or imperfections within the economic systems—those that, if pursued, provide the potential for economic returns. Opportunity recognition within environmental entrepreneurship involves recognizing trends in eco-institutional convergence. For example, as governments continue to tighten regulations on the amount of greenhouse gas emissions that companies can produce, an opportunity lies in understanding what the regulations will mean to private industries, particularly those that are heavy polluters, and in creating a solution that allows them to lower greenhouse gas emissions through new technology or means of production. Helping industries to comply with upcoming regulations while making their operation more

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efficient and profitable is a win–win scenario. In short, environmental entrepreneurs can benefit from understanding how market institutions are changing for valuable natural and environmental resources like air, water, climate, and ecosystem services.

Opportunity creation refers to entrepreneurial actions that alter market systems to open up new opportunities for economic return. Opportunity creation in environmental entrepreneurship involves creating eco-institutional convergence by intervening in the system. In other words, the entrepreneur is proactively engaged in altering market institutions in a way that aligns ecological goals or outcomes with economic return. Entrepreneurs can create new market institutions and incentive systems that enable profitability for more environmentally friendly business strategies, models, technologies, and products. While there is no single right way to go about creating opportunities, there are indeed strategies to explore when developing a sustainable venture.

The strategies for opportunity recognition and creation in environmental entrepreneurship are varied, but recent history suggests that many strategies have proven successful. The key to understanding how to see or unlock emerging opportunities for environmental entrepreneurship rests in understanding the existing barriers to efficient markets.² In other words, the key to environmental entrepreneurship rests in understanding market failures in the current economic system. Later, we review the five institutional strategies that entrepreneurs can pursue to resolve the various types of market failure. We also distinguish between the institutional strategies and general business strategies, because institutional strategies alter institutional frameworks to enable sustainable ventures by addressing the market failures found within the system.

These institutional strategies are summarized in Table 3.1.

STRATEGY #1: DEFINE AND ENFORCE PROPERTY RIGHTS

The *property rights entrepreneur* implements new means of defining or enforcing property rights in order to enable a profitable venture.^{3, 4} A *property right* is the authority to determine how a resource or asset is used. Property rights can apply to real estate (like land and its improvements), products (like cars and TVs), natural resources (like underground minerals), or intangible assets (like various forms intellectual property).

The purchase of a product, or other asset, is perhaps best conceived of as a purchase of a bundle rights, rather than as a purchase of a physical asset,⁵ because physical assets and products come with a set of rights associated with them. Those rights may be broad and encompassing, or they may be constrained by legal structures or contracts. For example, rights to the minerals under housing subdivisions (i.e., oil or gold) are often not included with the right to occupy and build a house on the land. In subsidized housing, property buyers often relinquish their right to resell their property at market rates in return for a lower price when the house is purchased. Property rights may include a mix of the rights to (1) *possess* an asset,

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TABLE 3.1 Institutional Strategies for Sustainable Venturing

<i>Institutional Strategy</i>	<i>Entrepreneur</i>	<i>Action</i>	<i>Example</i>
1. Define and Enforce Property Rights	Property rights entrepreneur	Establish excludability for asset, product, or service	Cap-and-Trade systems for greenhouse gas emissions
2. Internalize Externalities	Institutional Entrepreneur	Develop new economic institutions to eliminate externalities	Carbon taxes, regulations, and voluntary industry standards
3. Develop Informational Structures	Informational Entrepreneur	Develop systems to enhance information in markets	Green certification and labeling systems (Energy Star, organic foods, green building)
4. Transform Government Intervention	Political Entrepreneur	Motivate changes to taxes, subsidies, or other government incentives	Renewable energy feed-in tariffs and production tax credits Elimination of fossil fuel subsidies
5. Break Monopoly Power	Market-Appropriating Entrepreneur	Destroy monopolistic positions through new technology or deregulation	Deregulation of telecommunications industry, Net metering policies for electric utilities

(2) *utilize* an asset, (3) *exclude* others from its use, (4) *obtain a stream of income* from the asset, and/or 5) *transfer* (sell) the rights to others.^{6–8}

Rights may be owned by states (*public property*) or by private individuals or organizations (*private property*). *Common property* refers to property that is held in common by a group or subset of individuals or organizations. Certain assets or resources may also not be subject to any ownership, with property rights to them undetermined or nonexistent. Property without defined rights is known as *open-access*, since use is open to all. Wildlife, for example, is often considered as being unowned. Property in the *public domain* may be seen as open-access, since all members of society may use it. The writings of Shakespeare, for example, exist in the public domain and may be copied and used by members of a society. In contrast, writings of more recent authors may be copyrighted and therefore exist as private property. Finally, property rights may be informal, meaning that they are not codified in law, contract, or written title, but are nonetheless recognized by certain groups.⁹

Good property rights regimes tend to make markets work effectively. For example, copyright law allows authors and musicians to benefit economically from their work, and therefore encourages the development of new works that provide entertainment and insight. Property rights work in markets because they provide *excludability*, meaning that owners of an asset can prevent others from using it (i.e., those who do not pay to use it).¹⁰ In contrast, poorly defined or enforced property rights regimes can cause markets to fail or to operate suboptimally. The inability to prevent users from copying movies, for example, can

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reduce marginal returns to studios and diminish incentives for the production of new movies. The inability to *enforce* property rights can result in similar outcomes, rendering property rights effectively nonexistent. With movies, for example, the ability of consumers to make illegal digital copies without penalty may result in the same outcome as the lack of copyright laws: reduced returns and diminished incentives for production.

Poorly defined or enforced property rights regimes for natural and environmental resources are arguably one of the root causes of environmental degradation.¹¹ Compared to property rights for products and capital assets, property rights regimes for air, water, and other natural capital are often underdeveloped. The best example of a resource with little property right protection may be the high seas. Under the United Nations' Law of the Sea, individual countries maintain territorial waters and an exclusive economic zone, which extends up to 212 miles beyond their shore. Each country has an exclusive right to use these zones for various economic purposes, including tourism, fishing, oil production, or wind-power generation. International policy prevents foreign countries from exploiting economic resources in these zones, and individual nations who own them can implement policies to preserve the resources within them. Beyond this 212-mile limit, however, no particular individual or nation owns the high seas. While some constraints exist in international agreements, the high seas have a tendency to be over-fished because there is little regulation, and individual fishers have little motivation to leave breeding stocks in place (they likely assume that if they don't harvest the fish, someone else will). In short, the outcome of the lack of ownership of resources on the high seas can be detrimental to important economic resources, like fisheries.

This result of poorly defined or enforced property rights is popularly known as the *tragedy of the commons*,¹² a concept briefly mentioned in the previous chapter. The tragedy of the commons may result when a resource is unowned and access to it is unconstrained.¹³ The analogy typically used to illustrate the tragedy of the commons is that of a common cattle pasture in villages. Each villager who has access to the commons is motivated to place more cattle on the pasture, because doing so increases his or her returns. But this occurs at the expense of the common good because, as each villager adds more cattle, the grass may die, and the land's ability to support cattle degrades. The tragedy of the commons occurs because no mechanism exists for excludability, and because the commons is *rivalrous*. Rivalry refers to the fact that one person's use of a product or asset diminishes or precludes another person's ability to use that same product or asset. The meaning of rivalry becomes more obvious when one thinks about *nonrival goods*, such as a digital musical recording, which can be listened to by numerous individuals in multiple locations without diminishing the quality of the recording or the ability of any of them to listen to it.

Goods that are nonexcludable and rivalrous (as in the tragedy of commons) are often referred to as *common goods* (despite the fact that it is their open-access [not common ownership] that creates the problem). Goods that are nonexcludable and nonrivalrous are known as *public goods*. Many people have historically

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referred to the air as a public good. While this is probably no longer the case (due to large-scale air pollution), air used to be considered a public good because individuals largely cannot be excluded from breathing it, and one individual's breathing does not materially diminish another person's ability to breathe. A *congestible good* is one that is nonrival at low levels of use, but becomes rivalrous at higher levels of use. Typical examples of congestible goods are national parks and highways. When highways are largely empty, adding another vehicle does not diminish the experience for other drivers. But when highways are congested, adding another vehicle can impact the speed of the traffic and diminish the experience of other drivers. Likewise, when national parks are relatively empty, the presence of other tourists has little effect on the aesthetics of the experience. But as the park becomes more crowded, the aesthetic experience can decline substantially from traffic congestion and noise.¹⁴

Nonexcludability in rivalrous goods creates two primary economic problems. The first is that nonexcludability can result in overuse (as in the tragedy of the commons analogy). The American bison, which roamed North America in millions, nearly became extinct because excludability was not established through ownership or other means. The second problem is that entrepreneurs will be less motivated to produce goods for which they cannot exclude nonpayers. Intellectual property right laws (patents and copyrights) were developed to change this dynamic and provide excludability for innovators and artists, which increases returns to valued works and innovations, and increases innovation.

Environmental entrepreneurs should pay special attention to the nature of property rights and the excludability of resources and or products because changes in the property rights system and the excludability of resources can create economic opportunities. Entrepreneurs who pursue opportunities related to property rights are termed *property rights entrepreneurs*. Property rights entrepreneurs establish excludability for open-access goods through the development and enforcement of property rights regimes. Property rights entrepreneurship can be accomplished in a number of ways.¹⁵ *Political strategies* involve collaborating with government agencies to create new property rights regimes or to increase enforcement of existing ones. In one classic example, lighthouse keepers established excludability for their services by gaining the right to charge ships for lighthouse services at the local port.¹⁶ Without the right to charge at port, lighthouse keepers would have had a difficult time excluding nonpayers from using the lighthouse. In Africa, elephant preservation has been aided by the assignment of herd ownership to local tribesmen.¹⁷ In the western United States, the development of property rights regimes for water has helped, albeit imperfectly, allocate scarce water and encourage agricultural development. In fishing, producers have banded together to create transferable quota systems that allow a more sustainable harvest while preserving fishing rights.

The expansion of property rights can create opportunities beyond the environmental. For example, giving people real property rights through the enforcement of land titles can create game-changing opportunities in developing countries. As Hernando de Soto points out, many people in the developing

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world own assets, “but they lack the process to represent their property and create capital. They have houses but not titles, crops but not deeds, businesses but not statutes of incorporation.”¹⁸ The property rights entrepreneur seeks to alter the institutional environment such that property rights are defined and enforced, which allows for the development of opportunities within the system.

We can see how imperative a well-functioning property rights system is to the economic health of a society. In recent years there have been some interesting and groundbreaking developments in property rights regimes. A particularly interesting example of changes within property rights regimes is the idea of extended producer responsibility. *Extended producer responsibility*, or take-back, laws intend to reduce waste by altering property right systems. The basic idea is that producers are responsible for a product at the end of its useful life. In essence, ownership reverts to the producer after the useful life of the product. Because the producer must take back the product, it is expected that the price of the product will reflect its relative disposal costs, and producers will be motivated to reduce that cost in order to make their product more competitive. While not without their limitations, approaches that extend producer responsibility can provide advantage to entrepreneurial firms that figure out how to design their products for recycling, remanufacturing, disassembly, or lower cost handling and disposal.

As air pollution and the recognition of the potential for climate change has increased, it has become more obvious that air is a scarce and rivalrous good, and that property rights for air are important to the proper allocation of air resources, at least as a disposal mechanism for our waste. While most efforts to reduce pressure on air resources have been regulatory, a number of property rights regimes have also emerged. These have become popularly known as *cap-and-trade* regimes. The basic idea of a cap-and-trade regime is that property rights for air emissions are created, typically as an emissions permit. Each permit allows a certain level of emissions, and scarcity is created because only a limited number of permits are allowed. Permits are either distributed to existing firms, or sold by the government at auction. Once distributed, permits may be traded on the open market in order to encourage their allocation to the highest valued use.

Cap-and-trade systems were first used in the United States for the management of sulfur dioxide emissions in electricity generation, and they have been largely effective in reducing these emissions efficiently. A number of cap-and-trade regimes for carbon emissions exist worldwide. The foremost carbon cap-and-trade systems are operated under the Kyoto Protocol, signed by 191 countries. Cap-and-trade systems create commodities markets for emission permits and incent emission reductions because unused permits may be sold in these markets. Cap-and-trade regimes allow companies that can reduce emissions inexpensively to sell their rights to companies whose pollution control costs are higher. These systems also encourage technologies and processes that are less polluting because businesses with fewer emissions may sell, or do not need to purchase, emissions permits.

Technological strategies may also be used to enhance property rights regimes, establish excludability for a good, and create new opportunities.¹⁹ The technique

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of branding cattle on the open range helped establish ownership and prevented the cattle from being stolen. The invention of barbed wire took that one step further and allowed landowners to keep their cattle on their land and cattle owned by others out. Technologies to prevent digital copying of music and videos enable excludability as well as artists' ability to choose whether or not they wish to freely share or sell their creations.

Motivating political action or developing technology to establish new property rights regimes is not a simple task. Political strategies often require collective action on the part of a group of entrepreneurs with shared interests in a particular outcome. Technological strategies require new thinking about how markets work and how enforcing new property rights can alter incentives. But perhaps most importantly, property rights regimes tend to evolve naturally as resources become scarce, and value can be captured in their preservation or efficient use. Understanding trends in property rights regimes for environmental and natural resources, and even contributing to their development, is a way for entrepreneurs to capture and create opportunities for more sustainable business models.

STRATEGY #2: INTERNALIZE EXTERNALITIES

In Chapter 2, we learned that the economic activity of one party can place costs (or benefits) on other parties. We referred to these costs (benefits) as externalities, and referred to them as side effects of one party's activity on another party for which no compensation is provided. These side effects can be positive or negative, and can be referred to as *external benefits* or *external costs*. That is, these effects may provide benefits to or incur costs on parties external to a transaction or economic activity. Externalities are also referred to as *spillover effects*, since they overflow onto parties not involved in the activity or exchange. The terms *positive externality* and *negative externality* are also used to indicate whether spillover effects are beneficial or detrimental to affected parties. The *institutional entrepreneur* develops new institutions that internalize externalities, and thereby enables more environmentally or socially sustainable ventures.²⁰

Both positive and negative externalities are common in economic systems. Positive externalities, for example, may occur in the practice of beekeeping. While beekeepers traditionally keep bees for the harvesting of honey, bees also serve to pollinate orchards and crops. If the bees pollinate neighboring crops, the beekeeper is providing direct benefit to neighboring farms. Because it is difficult (or costly) to track such pollination services, it can be difficult for beekeepers to get paid for pollination services. Positive externalities may also exist in real estate, especially in the case of retail stores. A large department store attracts customers, who may stop by a small eatery after shopping. Thus, the owner of the eatery next to such a store benefits from the increased traffic and commerce the department store creates. A similar dynamic occurs with golf courses and ski areas, where much of the benefit can accrue to holders of real estate surrounding golf courses or ski slopes.

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One of the more interesting types of positive externalities is known as a *knowledge spillover*, which is often related to economic development. The term refers to knowledge, inventions, and technologies that cross organizational boundaries and benefit other companies than those that developed the knowledge. Knowledge spillovers occur through various means, but the movement of employees from one firm to another is a primary factor; such spillovers are one reason why technology ventures tend to cluster around universities and in regions like Silicon Valley.

In the previous chapter, we discussed how the burning of coal for electricity generation could create negative externalities because of emissions. Driving a gasoline-powered vehicle also releases emissions that, when combined with thousands of other vehicles, can create smog that has harmful health impacts. Scientists argue that the burning of fossil fuels, whether coal, oil, or natural gas, produces carbon emissions that create climate change and have associated impacts on ecosystems, land productivity, water supplies, and sea-level rise. Negative externalities also often arise in real estate. When an owner allows a property to fall into disrepair, it can substantially reduce the value of neighboring properties. A loud muffler or late-night party can impinge upon others' enjoyment of their environment, and runoff from pesticide and fertilizer use can impact fisheries and water quality downstream of agricultural lands.

Whether positive or negative, externalities can influence market outcomes, and institutional entrepreneurs are well aware of this. The inability to capture external benefits or to force others to pay their full external costs causes market perversions and can reduce entrepreneurial action in certain sectors. Most importantly, externalities can cause some entrepreneurial endeavors to be less financially attractive and others to be overly financially attractive. Firms producing substantive positive externalities are unable to capture as much value as they create. In short, it may be argued that organizations producing external benefits are underpaid for their products or services. In some instances, this means that the business is still viable; it is just not as competitive as it might be relative to alternatives. In other instances, the inability to appropriate the value created means the business is not at all viable. In contrast, firms creating substantive negative externalities are getting a bit of a free ride at the expense of others. In a market system, this makes them relatively more competitive.

Internalizing externalities refers to the process of eliminating externalities from market systems.²¹ Entrepreneurs may gain from the internalization of either negative or positive externalities. When negative externalities exist, entrepreneurs in competing businesses or industries may figure out new ways of making individuals and organizations that create external costs pay those costs (forcing others to internalize their negative external costs). When positive externalities exist, entrepreneurs may figure out new ways to capture the value of the external benefits (in essence, preventing the dissipation of the benefits that the company creates).

Broadly speaking, internalizing externalities is achieved through the creation of new economic institutions, and the institutional entrepreneur is engaged in

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the process of developing new institutions that internalize externalities, whether positive or negative. The classic example, which we referred to earlier, is the patent system, which enables individuals and companies to benefit from their innovations. Many think that internalizing externalities requires government action. But entrepreneurs often find creative ways of capturing positive externalities or ensuring that competitors bear external costs.

Entrepreneurs can appropriate the value of positive externalities through innovative business models. Beekeepers, for example, eventually discovered that they could capture the value of pollination by transporting hives to farms in need of it. Because the hives can be moved on trucks, beekeepers are able to charge farms for pollination services while they benefit from the sale of honey produced.²² The concept of a shopping mall may also be seen as a way to internalize the positive externalities of a retail location. The shopping mall rents to both the large “anchor stores” that attract customers and to the small eateries and specialty shops. Anchor stores typically pay less rent per square foot of space, effectively transferring some of the benefit created by the anchor store.²³ Golf courses are increasingly developed in concert with the surrounding residential lots, enabling the developer to capture the value created by the course for the surrounding real estate.

These methods may be referred to as *integration strategies* because they integrate different businesses in response to the externalities that may exist between them. When one business activity creates substantive positive externalities for another, integration strategies can be used to increase profitability of the business creating those benefits, by buying, or developing, both businesses at the same time. Another example of a positive externality can be farming mussels in nutrient rich waters. Seawater that is overly dense in nutrients can cause fish to suffocate, which is a major environmental concern. Mussel farming in nutrient dense water has been shown to be a viable way to reduce excess nutrients in coastal water while producing a high-value seafood product at the same time.²⁴

Entrepreneurs can also benefit by forcing competitors to internalize negative externalities. This may be achieved in a variety of ways. One way is through the development of taxes that require producers to pay a sum equivalent to the negative externalities they produce. These are referred to as *Pigouvian taxes*, named after the economist, Arthur Pigou, who first suggested taxes as a means of alleviating market effects of negative externalities. Theoretically, a Pigouvian tax forces polluters to internalize the costs of production, which raises prices for products that are harmful to the environment.²⁵ In such case, products that produce environmental damages will be more expensive for customers and be at a cost or price disadvantage relative to similar products that cause less harm. A Pigouvian tax, therefore, is designed to enable full-cost pricing. Note that this is not a tax used to repair environmental degradation, but rather one that internalizes the cost of the damage in the price of the good. Pigouvian taxes benefit entrepreneurs in industries with competing products that can be produced without similar harmful effects, and therefore with less tax.

Gasoline taxes are a type of Pigouvian tax because they force consumers to internalize some of the external costs of fuel use (such as the creation of smog).

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Carbon taxes (taxes levied on carbon emissions of various sorts) are another (and related) type of Pigouvian tax. Both could make entrepreneurial activity in renewable energy industries more attractive because the higher cost of fossil fuels makes renewable energy options more competitive in the marketplace.

While Pigouvian taxes may be useful for solving many externality problems, they are far from a perfect solution. There are three main challenges posed by Pigouvian taxes. The first is the challenge of calculating the correct level of tax to apply. It is hard to agree on what level of pollution one is willing to accept in relation to the price one is willing to pay for the good. If the tax is too high, then society's demand is not being met, and people's utility is not being maximized. On the other hand, if the tax is too low, then society will demand more of the good and too much pollution will occur, making society as a whole worse off. The second challenge to Pigouvian taxes is that they are subject to political manipulation, which may cause them to be set at a less than ideal level (the level that is equivalent to the external damages). The last challenge is that they have generally proven to be difficult to create because they have been politically unpopular in many countries, particularly in the United States. They nevertheless remain as a tool, or at the least a source of opportunity, for environmental entrepreneurs wishing to level the playing field for environmentally superior products and services.

Regulatory strategies may also be utilized to force producers of negative externalities to internalize costs. Often referred to as *command and control*, regulation requires producers to follow specific guidelines of behavior in the production process. *Performance standards* typically require producers of pollution to limit emissions to a certain level. *Prescriptive standards* require the utilization of specific technologies or processes for pollution abatement. Both performance and prescriptive standards are in popular use throughout the world to control emissions into air, water, and soil. As with taxes, such regulations can aid environmental entrepreneurs by limiting the negative externalities created by competing producers. Dupont, for example, was argued to have benefited from the ban on chlorofluorocarbons (used as refrigerants), which depleted the ozone layer in the upper atmosphere. Scientists realized that commonly used chlorofluorocarbons were destroying the ozone layer and allowing cancer-causing radiation to reach the earth's surface. After analyzing the scientific data, Dupont came out in support of the ban and committed to ceasing the production of chlorofluorocarbons. In the 1980s, politicians in developed countries agreed to the Montreal Protocol, which banned the production of the primary ozone-depleting chemicals. Dupont, had taken a bold stance, but also had developed chemical alternatives, which it then offered to the marketplace.

Voluntary industry initiatives may also be useful, particularly among firms with superior environmental technologies or products. Such initiatives often take the form of standard setting, wherein stakeholders agree on certain rules of behavior and producers decide whether they wish to follow those standards. Those who do may benefit from the ability to market their compliance to their customer base. The Forest Stewardship Council (FSC), for example, certifies lumber to ten principles of responsible forest management. The Home Depot,

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a retailer of hardware and lumber products now carries FSC certified lumber. The International Organization for Standardization (ISO) 14000 environmental management standards provides another example of stakeholder generated standards. The ISO 14000 standards are management standards that prescribe systems for governing the environmental aspects of an organization's operations. A facility or company can be certified to the ISO 14000 standards and use that certification to communicate to external stakeholders that they are managing their environmental performance.

One of the best ways to internalize externalities is through property rights strategies, which we discussed in the prior section.²⁶ Indeed, many economists suggest that all externality problems are, in essence, property rights problems. But externalities may also be understood as a transaction cost problem. *Transactions costs* refer to the costs of exchange as opposed to the cost of production and include the costs of searching for information on products, negotiation, contracting, and enforcement of contracts. Where transaction costs are higher than the value of an exchange, it is possible that the exchange is not completed, leading to a suboptimal economic outcome and an opportunity left on the table. If indeed the goal of the transaction is to make both parties better off, then both parties lose an opportunity to increase their well-being. For the institutional entrepreneur, strategies to reduce transaction costs are perhaps the most effective in correcting for market failures and creating an opportunity for a sustainable venture.

Technology also plays a major role in lowering the cost of gathering information. Moreover, understanding the institutional environment in which one is participating is equally imperative, especially in the pursuit of sustainable venture opportunities. This is because the legal and regulatory environment can differ from country to country. Having the ability to enforce contracts and other legally binding agreements will allow parties to more swiftly come to terms. If negotiations and contract enforcement take too long or are uncertain, the agreement may be tough to finalize. Institutional entrepreneurs try to build certainty into their agreements and tend to operate in legal environments that afford them retribution if the other party does not uphold their end of the bargain.

As we can see, there are multiple opportunities for institutional entrepreneurs to exploit if they know where to look. By internalizing externalities through regulatory strategies, voluntary industry initiatives or by lowering transaction costs, entrepreneurs can develop and act upon entrepreneurial opportunities, which results in the possibility of generating profit while providing societal or environmental value.

STRATEGY #3: DEVELOP INFORMATIONAL STRUCTURES

The *informational entrepreneur* develops new market information systems that provide traction in markets and make them more efficient. Informational entrepreneurs have been prevalent in the market for environmentally oriented products and services. Indeed, informational entrepreneurs have been active in

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markets as diverse as organic foods, green buildings, renewable energy, energy efficiency, and sustainable forestry. The key to the success of these programs has been in creating systems that provide valid information to customers regarding the green or social attributes of a product or service. These informational entrepreneurs have resolved the common difficulty customers have had telling whether a product or service was actually more environmentally or socially friendly than other products.²⁷

Business strategists refer to this problem as *information asymmetry*.²⁸ Information asymmetry typically refers to the fact that sellers of products have access to superior information about the product or service than do buyers—that is, the available information is asymmetrical because one side has more of it than the other. Information asymmetry can diminish the efficiency of markets or even result in the failure of a market to exist at all.

Information asymmetry has been historically prevalent in markets for green products and services. The green building market is a good case in point. Not very long ago, it was very difficult for customers interested in the benefits of green buildings to distinguish between buildings that were truly green, and those that were not. This occurs because it is relatively easy for a builder to claim a building is green, when it is actually not, which creates a competitive dynamic in which those who are not authentic green builders have a cost advantage over those who are authentic.

In the United States, a certification system for green buildings, known as Leadership in Energy and Environmental Design (LEED), was implemented to solve the information problem. Created by the U.S. Green Building Council, the LEED program certifies new and existing buildings according to a set of efficiency, environmental, materials, and other criteria. Third parties certify buildings to help assure customers that the building meets LEED criteria. The system has helped provide traction to the market for green buildings and has spawned an industry of architects, consultants, builders, and other entrepreneurs looking to build more efficient and effective buildings.

Similar third-party certification and labeling systems for environmental products and services have had similar effects. Common examples include certification systems for energy efficiency in appliances (Energy Star), organic foods (U.S. Organic Foods Production Act), renewable energy credits (Green-e), fair trade (fair-trade coffee), and sustainable forestry (Forest Stewardship Council). While some of these are supported or sponsored by governments (i.e., the organic foods labeling system and Energy Star program), many have emerged as nonprofit organizations that support industry participants and others who share the goal of differentiating their products to customers. Most programs are associated with a label that may be attached to products that meet the certification standards, which is a way to communicate the product's attributes to the customer. Most certifications are also performed by a third party (not by the producer of the product) to help insure that the certification is authentic. After all, the credibility and success of these systems is dependent upon the validity of the criteria and the trustworthiness of their certification process. Debates of the

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nature of the specific criteria often follow, and systems to monitor the authenticity of certification are critical.

Regardless, certification and labeling systems for products with environmental or social aspects can help make markets for them work more effectively. By providing a means of communicating environmental and social benefits to consumers, they solve information asymmetry problems and help customers benefit from environmentally superior offerings. Most important, the emergence or creation of new, trustworthy certification and labeling systems represent an opportunity for environmental entrepreneurs to profit from the benefits they create.

Strong brands can have a similar informational effect, providing customers with some assurance that the companies' products or services meet expectations for superior environmental or social performance. Patagonia, for example, has built a brand that many customers affiliate with superior environmental performance not only by creating top-performing products with an "ironclad guarantee" for customer satisfaction, but also by making sure that their products are manufactured in the most environmentally and socially responsible manner. They are transparent in the facilities that they use in their global production, and they employ third-party auditors to ensure that their suppliers adhere to Patagonia's Social and Environmental Responsibility Guidelines. They also report their environmental stewardship measures on their webpage called the "Footprint Chronicles," which documents some of the environmental and social impacts of key products from design through shipping.²⁹

To the degree that customers care about the social or environmental attributes of their purchases, businesses can benefit by signaling their superior attributes through branding and certification programs, achieving potential competitive advantage in the marketplace. In Chapter 5, we will further address the nature of markets for social and environmental attributes, and discuss the means by which companies can successfully market those attributes.

STRATEGY #4: TRANSFORM GOVERNMENT INTERVENTION

Government intervention in markets is pervasive throughout the world. Intervention takes a variety of forms, including taxes, subsidies, and regulations. Whether one sees such intervention as a good or bad activity, the reality is that such intervention affects the competitive dynamics of markets. All too often, such intervention benefits entrenched interests with the power to influence political decisions. It may also benefit relatively dirty or harmful activities that are not in the best interest of society or ecological sustainability. Increasingly, however, governments are altering their perspective and providing greater support for environmental entrepreneurs. Motivating, or even just understanding, the nature of these changes can be a source of substantial opportunity for aspiring environmental entrepreneurs.

Political entrepreneurs motivate changes to taxes, subsidies or other government incentive structures to increase the competitive position of their companies and enable a more profitable venture.³⁰ Political entrepreneurship may be used

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by sustainable entrepreneurs to enhance the viability of a more social or environmentally sustainable venture. Political strategies can be quite effective for environmental entrepreneurs because the wind of popular opinion or ecological virtue may be at their back. In many cases, this support has allowed environmental entrepreneurs and associated stakeholders to emerge victorious over powerful and entrenched political interests. Often, the ability to revolutionize the nature of government intervention rests in the need to eliminate negative externalities, or more broadly speaking, to solve a social or environmental problem. It usually involves partnerships with stakeholders who have shared goals, which can create unusual, but sometimes productive, alliances.

Political changes and strategies have created substantive opportunities for renewable energy companies throughout the world. In Germany, the renewable energy sector has been buoyed by generous feed-in tariffs, which offer higher rates to generators using renewable technologies. In Japan, subsidies and a broad effort to decrease the island nation's reliance on imported energy helped make Sharp, Kyocera, and Sanyo some of the leading producers of solar cells. China has also recently been a supporter of renewable energy, and its companies now hold significant market share in solar cell production.

While these governments' decisions may be motivated by economic development and monetary gain, the result is a system that is friendlier to environmentally sustainable products and services. Regardless of the intent of the government, the political entrepreneur must be able to recognize the social or environmental impacts of all types of legal and regulatory decisions, as there may exist an opportunity to exploit the dynamic political system. In the United States, numerous states have passed renewable portfolio standards that require electricity producers to generate a certain percentage of their power from renewable sources. As of 2011, twenty-nine states had passed renewable portfolio standards, and the laws have spawned an industry of solar power installers in many of those states. U.S. policy has also strongly supported the biofuels industry, which created a boom in corn-based ethanol production, as well as significant investment into technology development for cellulosic ethanol and other biofuels. Most recently, the U.S. government has been stimulating the development of electric vehicle and associated battery technologies.

Efforts to alter the competitive position of environmental entrepreneurs may take a variety of forms, including taxation policy, regulatory policy, or more direct subsidies. Taxation policy can include tax credits or deductions for certain activities (often capital investments), or direct taxes. In the United States, the renewable energy production tax credit allowed those who install renewable generation capacity a tax credit of 2.2-cents/kilowatt hour of energy produced, for the first ten years of a facility's operation. The tax credit had a substantial impact on the installation of wind-power generation in the United States. Other tax credit programs support home insulation contractors, energy-efficient appliance manufacturers, and electric vehicle producers. As discussed earlier, direct taxes may also penalize polluting industries, which can benefit businesses with cleaner technologies or products.

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We would be remiss if we did not address a few caveats. Altering the nature of government intervention is no easy task. Success depends on the complex dynamics of public attitudes and political decision-making. In addition, promoting government intervention can create a number of unanticipated problems, both philosophical and practical. At a philosophical level, government intervention is often seen as counter to the functioning of a healthy free-market economy. Indeed, placing decision-making authority for the success of an industry in the hands of politicians can result in poor investments. A case in point may be the various policies that created the boom in corn-based ethanol production in the United States. While the policies were clearly successful in generating entrepreneurial action, many argue that the energy expended to produce these fuels is almost as great as the energy they are purported to conserve. In other cases, such as the renewable energy production tax credit, changes in the credit over time have created boom/bust cycles in the installation of wind turbines.

Yet, within the space of global competition, nations often find themselves competing against countries where substantive national investments in certain industries are made. The term *industrial policy* describes the extent and direction of government investment and support in certain industries or sectors. In some countries, such as Japan and Germany (and more recently the United States and China), industrial policy has strongly supported investment in industries such as renewable energy. Such subsidies can be utilized to help generate jobs and new ventures in important new areas, and it may be difficult to compete without them.

Unfortunately, government policy may also be utilized as a tool to redistribute wealth to certain sectors or individuals. *Rent-seeking* refers to efforts to manipulate the political process for individual or organizational gain without an associated benefit to the social good or the functioning of free markets.³¹ It is often difficult to untangle rent-seeking efforts from efforts to generate support for new and important industries that are in society's best interest. The goal of the environmental entrepreneur is to navigate these difficult issues to create sensible support for industries and sectors that have better ecological outcomes, but make economic sense. Typically, this means garnering support for industries that produce fewer negative externalities or generate greater positive ones. At the least, it seems that leveling the playing field by eliminating government subsidies for ecologically damaging activities should be a top priority of the environmental or political entrepreneur.

STRATEGY #5: BREAK MONOPOLY POWER

The *market-appropriating entrepreneur* breaks the monopoly power (or more generally market power) of incumbent firms in an industry.³² A *monopoly* means that there is only one supplier of a particular good or service. *Oligopoly* means that there are only a few suppliers of a particular product or service. In either case, competition may be reduced, and consumers can encounter higher prices from the lack of competition. Economists argue that monopolists are able to price their products above the competitive price and thereby achieve higher

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profits. Industries characterized by monopoly represent opportunities for entrepreneurs because of the potential to capture some of the excess profits within the industry. Entrepreneurs can appropriate some of the profit by eating into the monopoly power of an incumbent.

Natural monopolies exist when economies of scale are so large that there can be only one efficient producer of the product or service. In short, it is less expensive for one firm to produce all of the output required by the market. For many years, it was argued that electricity production and long-distance communications were natural monopolies and that the most efficient means of production was a single firm. Often, such industries are regulated in an attempt to insure that prices remain at a competitive level. Regulation may also result in a *statutory monopoly*, in which the status of the monopoly is assured by government policy. This is often the case with electrical power utilities subject to a regulatory authority that controls electricity prices so that the monopolist has the only legal right to serve a given geographical market. Some monopolies also exist as government-owned entities in which the state retains control over the providers' decisions.

Many argue that industries characterized by monopoly, oligopoly, or market power more generally tend to become technologically stagnant and shun the introduction of new technologies. This may be the result of the removal of competition, in which case companies lack the incentive to introduce new ideas or methods. Thus, another way to generate opportunity for environmentally friendly businesses is by breaking the monopoly power of stagnant industries. Because monopolistic industries have a tendency to not innovate, the persistence of monopoly power often dissuades technological advancement that is more environmentally friendly. In addition, monopolistic industries are an attractive target because they are often highly profitable.

Breaking monopoly power is, of course, difficult, because monopolists are often highly entrenched in their industries and are interested in defending their positions. But it may be accomplished in a number of ways. In the case of natural monopolies, the answer usually lies in technologies that operate efficiently at smaller scale. Nucor Steel and other “mini-mills,” for example, successfully competed against large integrated steel manufacturers by using innovative techniques for recycling steel into high-grade steel products. In the case of statutory monopolies, alternative providers need to make the case to politicians and legal authorities that they can successfully compete within an industry. MCI, an innovative telecommunications company, successfully made the case to authorities that the statutory monopoly over long-distance telecommunications in the United States was unjustified. The result was the break up of AT&T into a group of regional telecommunications providers and long-distance services. Their case was made more robust by the advent of microwave-based relay station technology, which eliminated the necessity of carrying signals over copper wires.

In the electricity generation industry, a number of developments have helped eliminate the strong monopoly power of utilities and increased opportunities for environmental entrepreneurs. The first has been the advent of technologies that allow electricity to be produced on a small scale at reasonable cost. *Distributed*

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generation refers to electricity production that occurs at small scale at or near its point of use. Distributed generation benefits from the fact that electricity does not have to be transmitted across large distances, but it typically results in higher costs of production compared to those of large centralized generating facilities. Distributed generation technologies include solar and wind power, as well as natural gas turbines.

The second development has been the advent of various regulatory changes that have forced utilities to accept electricity generated by other sources. Most U.S. states, for example, require utilities to accommodate customers who generate their own electricity. This is accomplished through *net metering* policies which allow customers with solar power or other electricity generation capacity to provide that capacity to the electrical grid when they are producing more than they use. Their electrical meter records the power provided to the grid and subtracts the amount from the total amount they use. The electricity customers only pay for the difference between what they use and what they produce. In other words, the landscape of power generation can be transformed from one large monopoly to many distributed sites that actually produce power for public consumption. Net metering laws have enabled electricity customers to install solar and wind generation and to benefit from their production when it exceeds their use. The laws have helped encourage new ventures in solar power system installation. A number of states have also deregulated their electricity utility industries, and customers can now choose their power providers. Some of these providers, such as Green Mountain Energy, offer renewable energy options to customers in deregulated states.

Key Learnings

In this chapter we discussed how an entrepreneur could help create opportunities for sustainable ventures. We drew upon the idea of exploiting market failures to develop institutional contexts that allow entrepreneurs to create new economic opportunities.

The first strategy for creating opportunity from a market failure is to define and enforce property rights. A property rights entrepreneur implements new means of defining or enforcing property rights in order to enable a profitable venture. Recall what happens when there are no property rights assigned to a good. The tragedy of the commons is a powerful example of what can happen when a group has unlimited access to use a good, but individual's are not responsible for its maintenance or the effects of their actions. Property rights entrepreneurs establish excludability for public and common goods through the development and enforcement of property rights regimes. Some political strategies involve collaborating with government agencies to create new property rights regimes or increase enforcement of existing ones. Cap-and-trade is an example of creating and enforcing property rights.

Technological strategies are also used to enhance property rights regimes, establish excludability for a good, and create new opportunities. Technological

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strategies may be easier to implement than political strategies, which may require a coordinated group effort. Technological strategies require new thinking about how markets work and how enforcing new property rights can alter incentives in the market. A rudimentary example of a technological innovation is the use of branding to identify one's cattle on the open range and the use of barbed wire to delineate private property and to keep one's own livestock in and another's out.

The second strategy we have investigated is the internalization of externalities. Externalities are the side effects of one party's activity on another party for which no compensation is provided, and they can be either positive or negative. The institutional entrepreneur develops new institutions that internalize externalities, making the producers of external costs pay for those costs, or enabling those who produce external benefits to capture those benefits. Internalizing externalities may allow more environmentally or socially sustainable ventures. Generally, the internalization of externalities is achieved through the creation of new economic institutions. While this may require government intervention in some instances, clever entrepreneurs find innovative ways to appropriate the value of positive externalities through creative business models. Entrepreneurs can also benefit by forcing competitors to internalize negative externalities. We have discussed one way to do this through the creation of Pigouvian taxes, which are taxes that require the producers to pay a sum equivalent to the negative externalities they produce. Regulatory strategies and voluntary industry initiatives are other approaches an entrepreneur can use to create opportunities via the internalization of externalities.

The third strategy we have explored is the development of informational structures—a strategy employed by the informational entrepreneur, who seeks to rectify information asymmetry and develops new market information systems that provide traction in markets to make them more efficient. We see informational entrepreneurs most often in the market for environmentally oriented products and services, like organic foods, green buildings, renewable energy, energy efficiency, and sustainable forestry. The key to the success of these programs has been creating systems that provide valid information to customers regarding the green attributes of a product or service. Examples of successful development of informational structures include the certified organic food program and the LEED certified building program.

The fourth strategy that entrepreneurs can use to create opportunity for a sustainable venture is to transform government intervention. Government intervention in markets is pervasive throughout the world and can include taxes, subsidies, and regulations; it also skews the free market, and all too often, such intervention benefits entrenched interests with the power to influence political decisions. Political entrepreneurs motivate changes to taxes subsidies or other government incentive structures to make their individual or corporate position more competitive. Political entrepreneurship is a strategy that environmental entrepreneurs may use to bolster the viability of their social or environmentally sustainable venture.

The fifth and final strategy an entrepreneur can use to create opportunities by exploiting market failures is to break monopoly power. We call this approach to entrepreneurship *market-appropriating entrepreneurship*. A monopoly is the condition in which there is a sole supplier of a good or service; in the presence of a monopoly, many would argue, technological innovation becomes stagnant. One way for a market-appropriating entrepreneur to break monopoly power is to introduce game-changing technology and disruptive innovations that operate more effectively and efficiently at a smaller scale. Although this is indeed challenging, the rewards for an entrepreneur that innovates a strategy to break monopoly power can be substantial.

The bottom line is that for an entrepreneur who identifies market failures or suboptimally performing markets, specific strategies can be employed to create value and generate rewards within the economic system. In some cases this requires recognizing emerging changes in the system. In other instances this requires changing the system itself.

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